How the Bankruptcy Risk Predictor works

A plain-English guide to the data, the model, the scoring and the app

The whole project in one sentence

The app takes 20 financial ratios of a company, gives them to a machine-learning model that learned from 43,405 real company statements, and returns the chance that the company goes bankrupt 1 to 5 years later, together with the reasons.

43,405	64 → 20	5	0.94
company statements learned from	financial ratios (best 20 kept)	models, one for each horizon (1–5 years)	average ROC-AUC on unseen companies

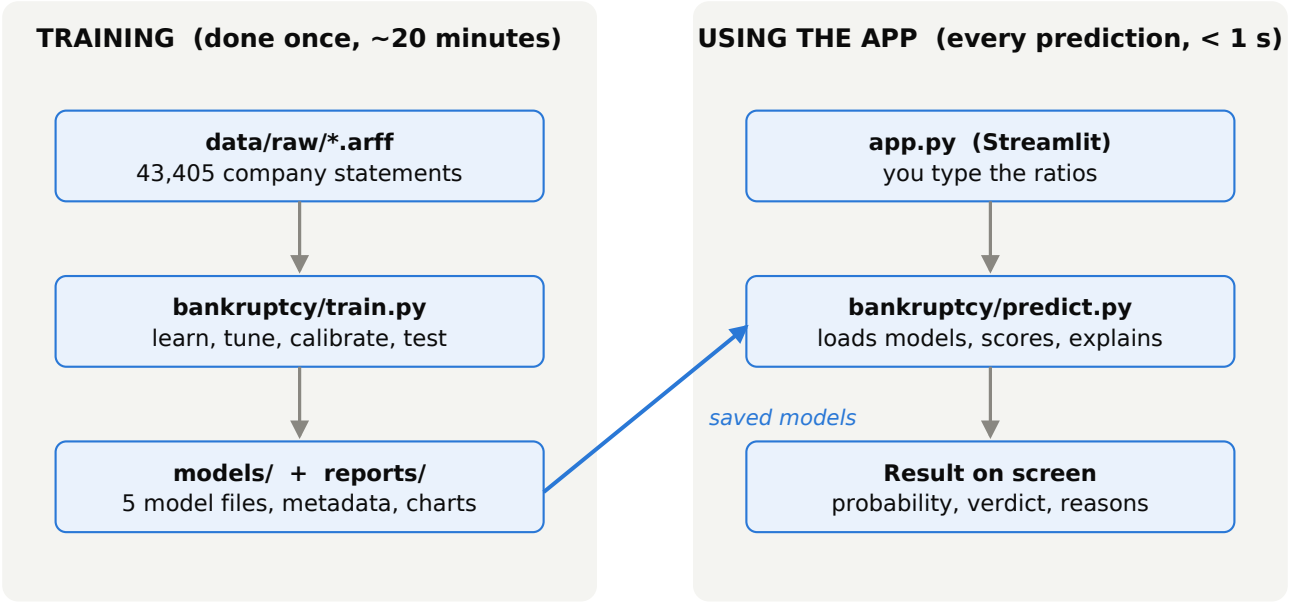
What's inside

- **1.** The big picture: how the pieces fit together
- **2.** The data: what the model learns from
- **3.** Training: how the model learns, step by step
- **4.** One prediction, worked through with real numbers
- **5.** Scoring: how we know the model is good
- **6.** What makes a company risky?
- **7.** What changed from the first version
- **8.** Using the app
- **9.** Limits, and a glossary

Model version 2.0 · trained 2026-10-17 · LightGBM gradient-boosted decision trees (LightGBM 4.7.0)

1. The big picture

Think of the project as two halves: a **factory** and a **shop**. The factory (training) runs once. It reads thousands of past company statements, finds the patterns that came before bankruptcies, and saves what it learned as five small model files. The shop (the app) uses those files every time someone asks about a company, and answers in a fraction of a second.



What happens when you press “Predict”

1. You enter the ratios
Up to 20 numbers such as “costs per unit of sales”. Unknown boxes can stay empty.

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2. The horizon picks the model
There is a separate model for 1, 2, 3, 4 and 5 years ahead.

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3. Hundreds of small decision trees vote
Each tree looks at a few ratios and adds or subtracts a little risk.

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4. The votes add up to a risk score
A single number: higher means more danger.

↓

5. The score becomes a probability
A calibrated formula turns it into a chance between 0% and 100%.

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6. The probability is compared with a threshold
At or above the threshold, the company is flagged as likely to go bankrupt.

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7. You see the verdict and the reasons
Which ratios pushed the risk up, and which pulled it down.

Where everything lives

Folder / file	What it is	In plain words
data/raw/	1year.arff … 5year.arff	The raw data: one file per horizon
bankruptcy/train.py	Training pipeline	The factory: builds and tests the models
bankruptcy/predict.py	Prediction engine	Loads the models, scores a company, explains why
bankruptcy/features.py	Ratio dictionary	Name and meaning of each of the 64 ratios
models/	model_1y.txt … model_5y.txt, metadata.json	The learned trees, plus thresholds, scores and examples
reports/	metrics.md, figures/	Test results and charts
app.py	Streamlit app	The shop: the web page you use
tests/	Automated checks	Make sure predictions and the app still work
docs/	This guide, deployment guide	Documentation

2. The data

The models learn from the **Polish Companies Bankruptcy** dataset, published by the UCI Machine Learning Repository (Tomczak, 2016). It was collected from the EMIS database of emerging-market companies. Bankrupt companies were analysed over 2000–2012, and companies that kept operating over 2007–2013.

One row = one company's financial statement, turned into 64 financial ratios, plus a label saying whether the company later went bankrupt. A **ratio** divides one number by another so that small and large companies can be compared. For example, *net profit ÷ total assets = 0.05* means the company earns 5 cents of profit for every 1 unit of assets it owns.

The 64 ratios fall into a few families

Family	The question it answers	Example ratio
Profitability	Is the company making money?	Net profit / total assets (Attr1)
Debt (leverage)	How much does it owe compared with what it owns?	Total liabilities / total assets (Attr2)
Liquidity	Can it pay the bills that are due soon?	(Current assets − inventory) / short-term liabilities (Attr46)
Debt service	Can its profits cover interest payments?	Operating profit / financial expenses (Attr27)
Efficiency	How well does it turn assets into sales?	Sales / total assets (Attr9)
Cost structure	Do costs eat up the sales?	Total costs / total sales (Attr58)
Size and growth	How big is it, and are sales growing?	Log of total assets (Attr29), sales growth (Attr21)

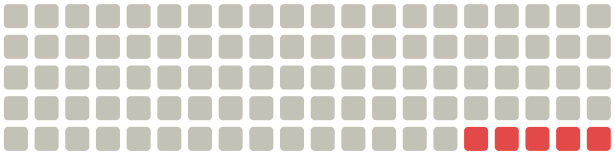
Five files, five horizons

The data comes in five files. Each one has a different gap between the financial statement and the bankruptcy outcome. The file names count the *year of the statement* within a five-year window, not the years ahead, so the numbers run in reverse:

Horizon	File	Companies	Went bankrupt	Share
1 year ahead	5year.arff	5,910	410	6.9%
2 years ahead	4year.arff	9,792	515	5.3%
3 years ahead	3year.arff	10,503	495	4.7%
4 years ahead	2year.arff	10,173	400	3.9%
5 years ahead	1year.arff	7,027	271	3.9%

Fixed in version 2: the first version labelled *1year.arff* as “1 year ahead”. According to the dataset's documentation it is the opposite: statements from year 1, with the bankruptcy status 5 years later.

Bankruptcy is rare



About 5 in every 100 companies
went bankrupt (4.8% of 43,405).

A lazy model that always says “safe” would be right 95% of the time, yet useless.

This imbalance is why the guide talks about *recall*, *precision* and *PR-AUC* rather than accuracy. Those scores focus on the bankrupt companies, which are the ones that matter.

Missing numbers

About 1.5% of the numbers are missing. LightGBM handles gaps natively. Interestingly, a missing value can itself be a clue: in the 1-year-ahead data, 96% of companies with no sales-growth figure went bankrupt, probably because failing companies stop reporting. The app therefore asks what an empty box means. **“I don't know the value”** (the default) fills in a typical value. **“Not reported in the accounts”** lets the model use what it learned about missing data.

3. Training: how the model learns

The same six steps run separately for each of the five horizons. Together they take about 15–20 minutes on a laptop (**python -m bankruptcy.train**).

Step 1. Lock away a final exam

Before anything else, 15% of the companies are set aside and not touched until the very end. They act like a final exam the model has never seen, so the test scores are honest. The split uses the same random seed as version 1, so both versions are graded on exactly the same companies.



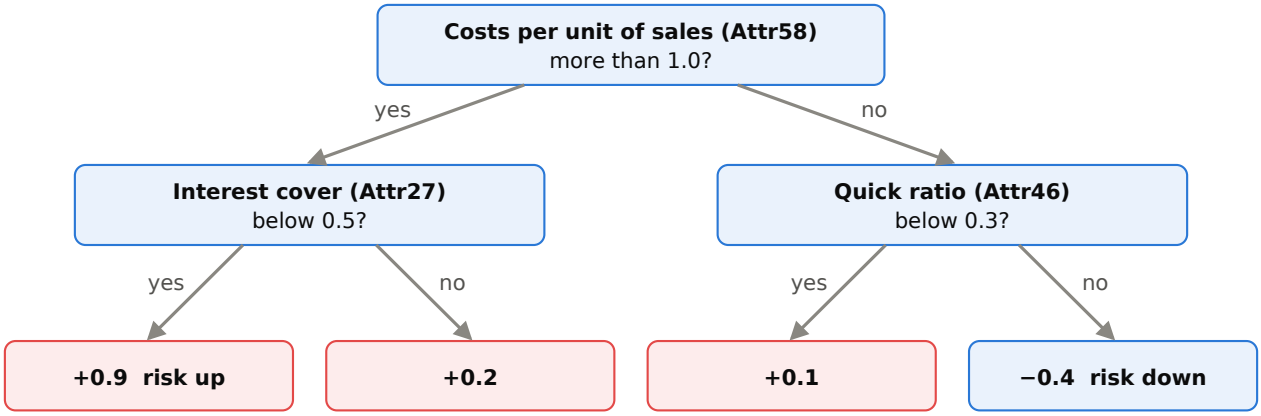
Both parts keep the same share of bankrupt companies (a “stratified” split).

Step 2. Pick the 20 most useful ratios

A quick model is trained on all 64 ratios using only the training companies, and we measure how much each ratio helped it separate bankrupt from healthy companies (its “gain”). The top 20 are kept. In tests, 20 ratios were just as accurate as all 64, and they keep the app's form short. 13 ratios made it into all five models, which shows they matter at every horizon.

Step 3. The model: LightGBM, a team of decision trees

A **decision tree** is a flowchart of yes/no questions about the ratios. At the end of each path is a small amount of risk to add or subtract. Here is a simplified example:

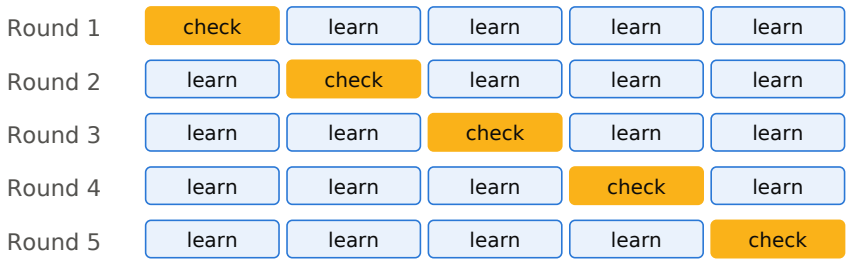


Simplified illustration. The real trees are learned from data and have 12–33 leaves each.

One tree on its own is weak. **Gradient boosting** builds trees one after another, and each new tree focuses on the mistakes the previous ones still make. The final risk score is the sum of all the trees' small answers. For example, the 1-year model uses 239 trees with up to 12 leaves each. **LightGBM** is a fast, popular library for this technique and is among the most accurate methods for table-shaped data like ours.

Step 4. Tune the settings with cross-validation

A tree model has settings: how fast it learns, how big each tree may grow, how many companies each leaf needs, and how strongly to avoid over-fitting. **Optuna** tries 40 combinations and learns which areas look promising. Each combination is judged with **5-fold cross-validation**: the training companies are cut into 5 parts, and the model learns on 4 and is checked on the 5th, five times over.



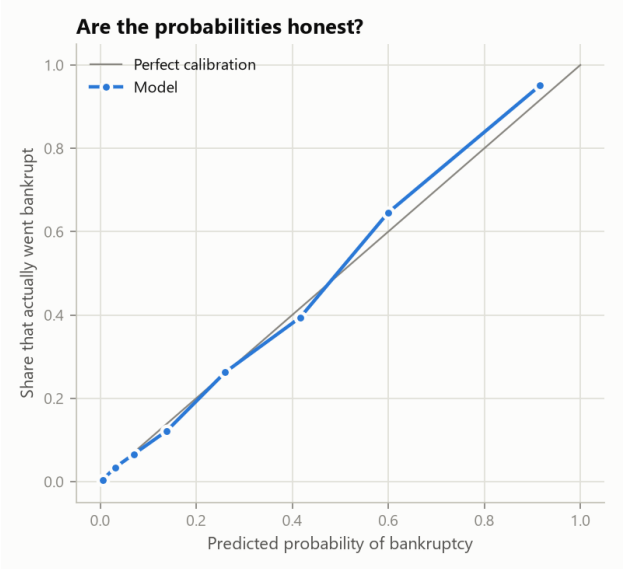
Each part is used once for checking, so every training company gets a fair, unseen prediction.

The combination with the best average *PR-AUC* (see section 5) wins. To avoid adding useless trees, training stops as soon as the checking part stops improving (“early stopping”).

Step 5. Make the probabilities honest, and set the threshold

Cross-validation gives every training company a prediction from a model that never saw it. These out-of-fold predictions are used twice:

- **Calibration.** A small formula (Platt scaling) adjusts the raw scores so that “30%” really means about 30 in 100 such companies went bankrupt. The chart below checks this.
- **Threshold.** Every cut-off from 2% to 95% is tried, and the one with the best F1 score (balance of catching bankruptcies and avoiding false alarms) is kept: 33% (1 yr), 29% (2 yr), 27% (3 yr), 24% (4 yr), 35% (5 yr).



Dots close to the grey diagonal mean the predicted probabilities match what really happened.

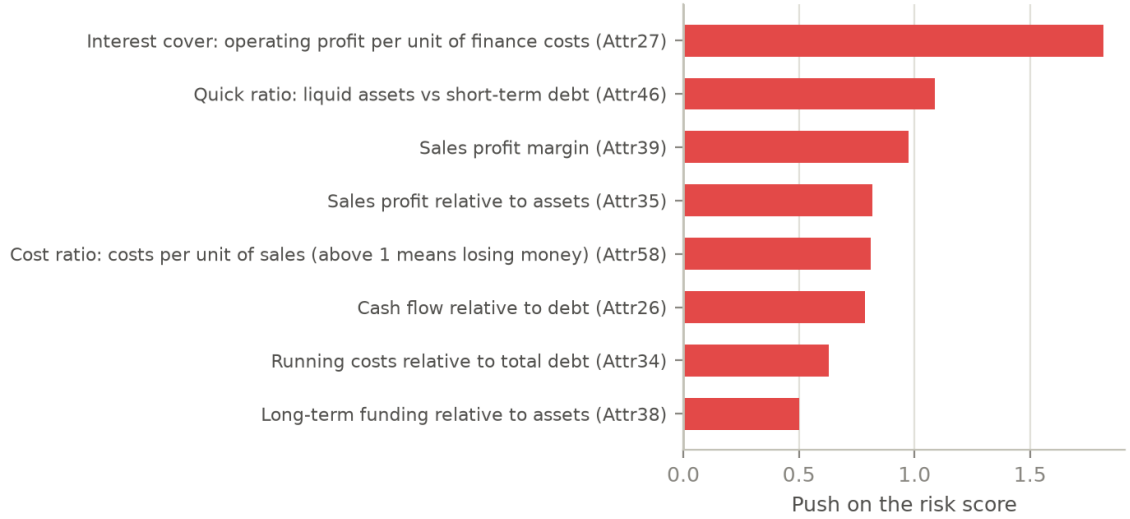
Step 6. Final exam

The final model is trained on all of the 85% training companies with the winning settings. It then predicts the 15% exam companies exactly once. Those are the scores reported in section 5.

4. One prediction, worked through

Let's follow a real company from the 1-year model's exam set (the app's “Distressed company” example). The model had never seen it, and the company really did go bankrupt. Here are the six ratios that influenced its result most:

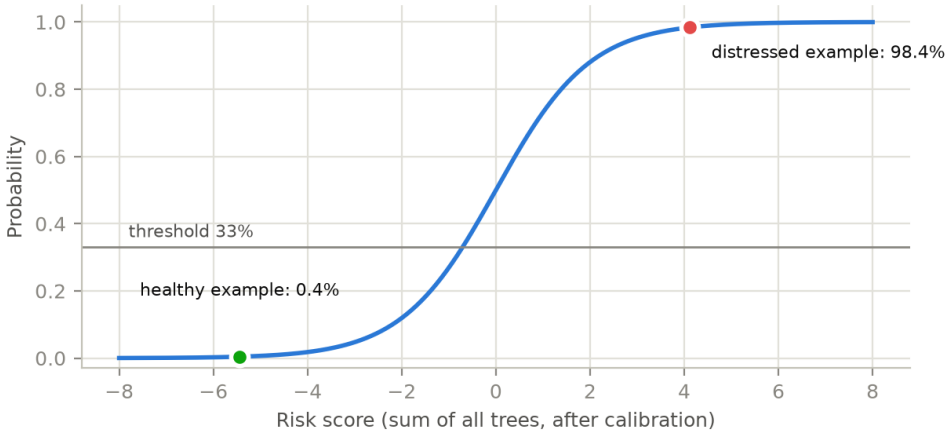
Ratio	This company	Typical company	Push on risk
Interest cover: operating profit per unit of finance costs (Attr27)	-10.192	0.996	+1.82
Quick ratio: liquid assets vs short-term debt (Attr46)	0.135	1.076	+1.09
Sales profit margin (Attr39)	-0.310	0.040	+0.97
Sales profit relative to assets (Attr35)	-0.385	0.062	+0.82
Cost ratio: costs per unit of sales (above 1 means losing money) (Attr58)	0.896	0.951	+0.81
Cash flow relative to debt (Attr26)	-0.189	0.214	+0.79



Red bars push towards bankruptcy, blue bars push away from it. These pushes are SHAP values, computed exactly by LightGBM for every prediction.

From pushes to a probability

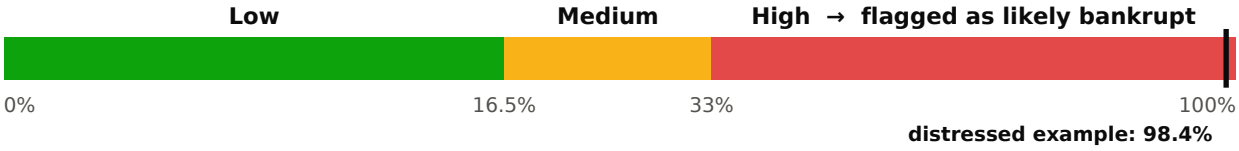
- **Start** from the score of an average company: **-4.60**.
- **Add** the pushes of all 20 ratios: **+8.71**. Risk score = **+4.11**.
- **Squash** the score into a probability with the formula $1 \div (1 + e^{-\text{score}})$. Any score becomes a number between 0% and 100%: big positive scores land near 100%, big negative ones near 0%. Here: **98.4%**.
- **Compare** with the 1-year threshold of 33%: 98.4% is above it, so the verdict is **likely to go bankrupt** and the risk level is **High**.



For contrast, the “Healthy company” example scores -5.44, which becomes a probability of 0.4%. That is far below the threshold, so it is rated Low risk.

Risk levels

Besides the yes/no verdict, the app shows a risk level. **High** means at or above the threshold. **Medium** means above half the threshold: not flagged, but worth watching. **Low** is everything below. For the 1-year model:



5. Scoring: how we know it is good

All scores come from the locked-away exam companies. Take the 1-year model: its exam had 887 companies, 62 of which really went bankrupt. Every exam company ends up in one of four boxes:

	Model said: bankrupt	**Model said: safe**
Really went bankrupt	**46** caught ✔	**16** missed ✘
Really survived	**18** false alarm ✘	**807** correctly safe ✔

Score	**What it means in plain words**	**1-year model**
Recall	Of the companies that really went bankrupt, how many did we catch? (46 of 62)	74%
Precision	Of the companies we flagged, how many really went bankrupt? (46 of 64)	72%
F1 score	One number that balances recall and precision (0 = useless, 1 = perfect)	0.73
ROC-AUC	Pick one bankrupt and one healthy company at random: how often does the bankrupt one get the higher risk score? 0.5 = coin flip, 1 = always.	0.96
PR-AUC	Average precision across all thresholds. A random guess would score about 0.07 (the bankruptcy rate); 1 = perfect.	0.82
Accuracy	Share of all companies classified correctly. Looks great, but a model that always says “safe” already gets 93%.	96%

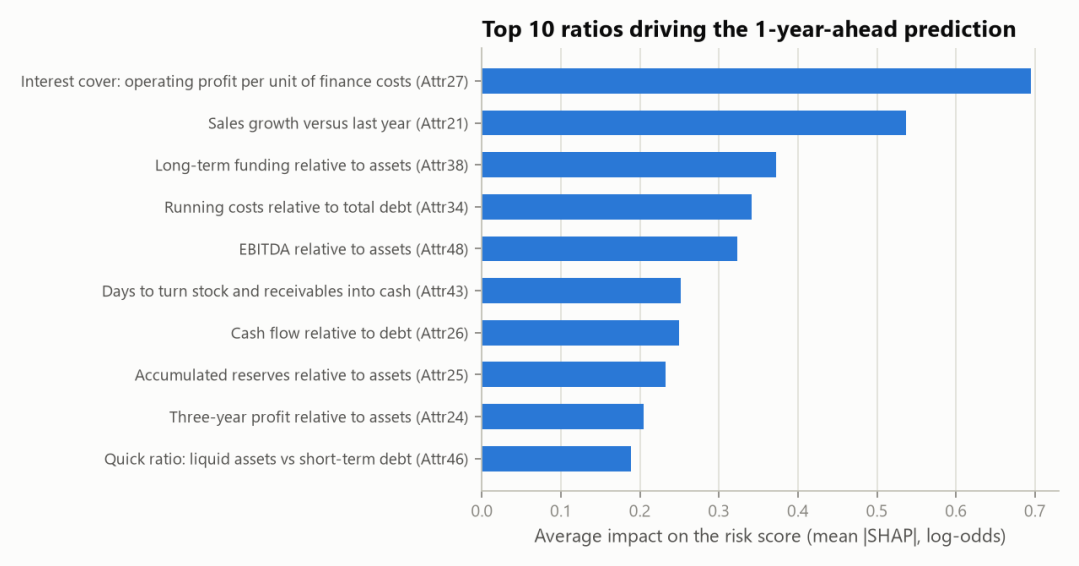
Results for all five horizons

Horizon	**Threshold**	**Recall**	**Precision**	**F1**	**ROC-AUC**	**PR-AUC**
1 year ahead	33%	74%	72%	0.73	0.96	0.82
2 years ahead	29%	64%	68%	0.66	0.93	0.72
3 years ahead	27%	54%	62%	0.58	0.93	0.63
4 years ahead	24%	60%	56%	0.58	0.93	0.67
5 years ahead	35%	73%	83%	0.78	0.98	0.85
Average		65%	68%	0.66	0.94	0.74

Each exam has only 40–80 bankrupt companies, so the scores carry some uncertainty. For example, the 1-year F1 of 0.73 could plausibly be anywhere from 0.64 to 0.81 (95% bootstrap range). Differences of a few points between horizons or versions should not be over-read. The ranges for every horizon are in reports/metrics.md.

6. What makes a company risky?

Averaging the SHAP pushes over all training companies shows which ratios each model relies on most. For the 1-year model:



Ratio	What it measures	What the data shows
Attr27	Interest cover: operating profit per unit of finance costs	**Low values are the danger sign** (lowest tenth 30% went bankrupt, middle 2%, highest tenth 3%)
Attr21	Sales growth versus last year	**Low values are the danger sign** (lowest tenth 22% went bankrupt, middle 4%, highest tenth 2%)
Attr38	Long-term funding relative to assets	**Low values are the danger sign** (lowest tenth 24% went bankrupt, middle 5%, highest tenth 4%)
Attr34	Running costs relative to total debt	Weak pattern on its own; useful in combination (lowest tenth 10% went bankrupt, middle 7%, highest tenth 6%)
Attr48	EBITDA relative to assets	**Low values are the danger sign** (lowest tenth 26% went bankrupt, middle 5%, highest tenth 4%)
Attr43	Days to turn stock and receivables into cash	**Both extremes are riskier** (lowest tenth 13% went bankrupt, middle 6%, highest tenth 12%)

Why a bar can look surprising: many ratios measure similar things (several of them are profit ratios, for example). The model may give the credit for a warning sign to one ratio and use a related one to fine-tune the result, so a single bar can sometimes point the “wrong” way. Read a company's reasons as a group rather than one bar at a time.

13 ratios appear in all five models (Attr6, Attr9, Attr21, Attr24, Attr25, Attr26, Attr27, Attr29, Attr34, Attr41, Attr46, Attr58, Attr60). They cover the classic warning signs analysts look for: weak interest cover, costs close to or above sales, thin liquidity, poor cash flow relative to debt, little accumulated profit and shrinking sales. The model's job is to weigh them against each other far more precisely than a rule of thumb can.

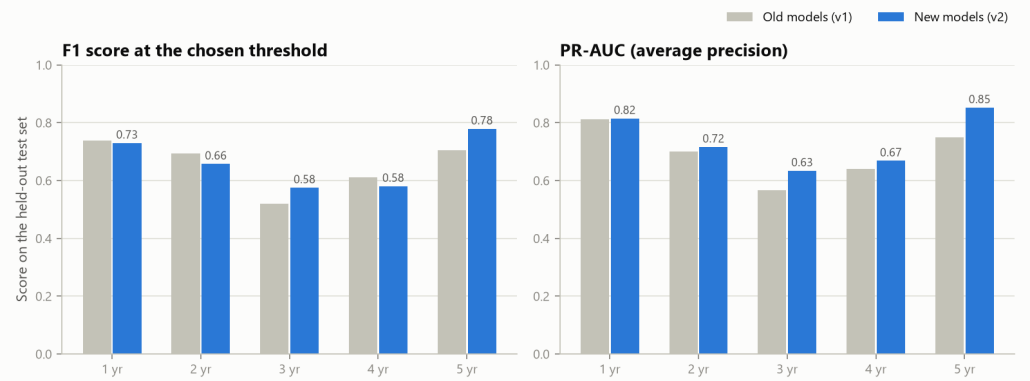
Charts for the other horizons are in reports/figures/ (importance_2y.png … importance_5y.png) and in the app's “How it works” tab.

7. What changed from the first version

Version 1 (the original notebook) was a good start. Version 2 keeps the same idea (LightGBM, one model per horizon, a tuned threshold) and fixes the issues below.

Problem in version 1	Fix in version 2	Why it matters
Horizons were reversed	Mapped as the dataset defines them	“1 year ahead” now really means 1 year ahead
The top ratios were chosen using all companies, exam included	Chosen from training companies only	Exam scores are honest (no “peeking”)
Settings set by hand; one of them (subsample) silently did nothing	40 settings searched with Optuna and 5-fold CV	Better, more reliable models
Threshold picked on one small validation split	Picked on out-of-fold predictions for all training companies	Threshold is stable, not luck
Probabilities distorted by class weighting	Calibrated with Platt scaling	“30%” means 30%
The app ignored the tuned thresholds	App uses each horizon's threshold	Verdicts match the tested model
No explanation of results	SHAP reasons for every prediction	You can see why
82 MB of version-locked pickle files	5 MB of plain LightGBM text files	Easy to deploy, safe to load
One notebook	Package, training command, 11 tests, Docker, deployment guide	Reproducible and maintainable

Scores on the same exam companies

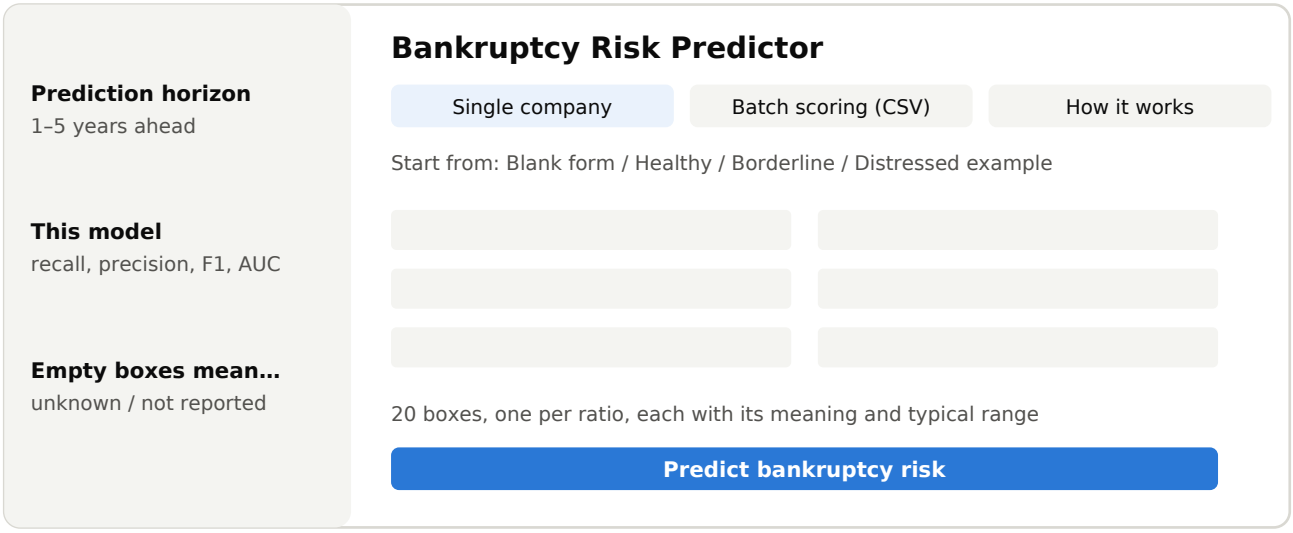


Average over 5 horizons	Version 1	Version 2
PR-AUC	0.694	0.737 (better on all 5)
ROC-AUC	0.930	0.944 (better or equal on all 5)
F1	0.654	0.665

F1 depends on one threshold and a small exam, so it moves around more: version 2 is higher on average but slightly lower on three horizons. PR-AUC and ROC-AUC look at all thresholds at once and improved across the board. Version 1 also had a small unfair advantage: its ratios were chosen with the exam companies in view.

8. Using the app

Start it with **streamlit run app.py**, or open the hosted link once it is deployed (see docs/DEPLOYMENT.md). The screen has a sidebar and three tabs:



Part	What you do there
Sidebar: horizon	Choose how many years ahead to predict. The form and the model change with it.
Sidebar: this model	See how well the chosen model did on its exam: recall, precision, F1 and ROC-AUC.
Sidebar: empty boxes	“I don't know the value” fills gaps with typical values (recommended). “Not reported in the accounts” treats them as genuinely missing, which the model reads as a warning sign.
Single company tab	Type the ratios, or load a real example company (healthy, borderline or distressed), then press Predict. You get the probability, threshold, risk level, verdict, and a chart of the reasons.
Batch scoring tab	Download the CSV template, fill one row per company, upload it, and get probabilities and verdicts for all five horizons in one table you can download.
How it works tab	A short version of this guide, the test results and the charts.

> **Tip:** hover over the small question-mark icon next to each box to see the ratio's exact formula and the typical range in the data. Values far outside that range are unusual and worth double-checking.

9. Limits, and a glossary

Use it wisely

- **Where the data comes from.** The models learned from Polish companies around 2000–2013. Companies in other countries, industries or economic periods may behave differently, so check results on your own data before relying on them.
- **Different collection periods.** Bankrupt and surviving companies were collected over different years. Part of what a model picks up may be differences between those periods rather than financial distress. This may be why some horizons score surprisingly well.
- **Same companies in several files.** A company can appear in more than one year file. That is why each horizon is modelled and tested separately. Mixing the files made scores look far better than they really were.
- **Small exams.** 40–80 bankrupt companies per horizon means a few points of uncertainty in every score.
- **A screening tool, not a verdict.** Use it to decide which companies deserve a closer look, and combine it with human judgement. It is not financial advice.

Glossary

Term	Meaning
Ratio / feature	One input number, e.g. total costs ÷ total sales.
Horizon	How many years ahead the model predicts (1–5).
Decision tree	A flowchart of yes/no questions that ends in a small amount of risk.
Gradient boosting / LightGBM	Many small trees built in sequence, each fixing earlier mistakes. LightGBM is the library used.
Training / test set	Companies the model learns from (85%) vs. the locked-away final exam (15%).
Cross-validation	Learning on 4 parts of the training data and checking on the 5th, five times, to judge settings fairly.
Hyper-parameters / tuning	The model's settings, and the search for the best ones (done with Optuna).
Early stopping	Stop adding trees once they no longer improve results on unseen data.
Risk score (log-odds)	The model's raw output. Higher means riskier, with no fixed range.
Calibration	Adjusting scores so the probabilities match real-world frequencies.
Threshold	The probability at or above which a company is flagged as likely bankrupt.
SHAP value	How much one ratio pushed one company's score up or down.
Recall / precision / F1	Share of bankruptcies caught / share of alarms that were right / their balance.
ROC-AUC / PR-AUC	Threshold-free scores of how well risky companies are ranked above safe ones.
Data leakage	Accidentally letting exam information into training, which makes scores look too good.